Systems Design

Document role

- This file owns the active subsystem baseline and cross-system modeling context for the build.
- Use this file to describe what the current system is, how the major subsystems fit together, and which assumptions are still active.
- Keep the final `48V` house/alternator architecture, shutdown order, and control-path lock in `docs/core/ELECTRICAL_48V_ARCHITECTURE.md`.
- Keep exact conductor, fuse, layout, and install-level detail in `docs/implementation/`.
- Keep decision/risk/open-question state in `docs/core/TRACKING.md` rather than letting this file become the live issue tracker.

Electrical and energy

Goals

- Deliver workday reliability with reserve margin
- Keep wiring safe, labeled, and serviceable

Electrical doc links

- Canonical `48V` architecture and alternator-control baseline: ELECTRICAL_48V_ARCHITECTURE
- Implementation topology (components, fuses, holders, gauges): ELECTRICAL_overview_diagram
- Fuse IDs, locations, housing methods, spares, and BOM mapping: ELECTRICAL_fuse_schedule
- Voltage architecture trade study (`12V` vs `48V`): ELECTRICAL_12V_vs_48V_trade_study
- Alternator architecture trade study history (research archive only; final decisions moved to the canonical `48V` architecture doc): ELECTRICAL_48V_dual_alternator_trade_study
- Solar option screening matrix (stringing + MPPT fit flags): SOLAR_configuration_matrix
- Electrical decisions, risks, and unresolved items: TRACKING

Planning snapshot (base model as-of `2026-07-19`)

- Battery bank: `3x 48V 100Ah LiFePO4` from BOM row 3 (`15.36 kWh` nominal at `51.2V` battery nominal).
- House architecture: `48V` core with Orion-Tr Smart `48V->12V` charging/step-down feeding a shared battery-backed `12V` junction.
- Inverter/charger: Victron MultiPlus-II `48/3000/35-50`, DC/inverter mode live-tested with no observed errors.
- Charge sources in current BOM: solar MPPT, dedicated `48V` secondary alternator path (`Mechman + WS500 + APM-48` migration baseline), shore AC charger path.
- Monitoring and protection: Cerbo GX, SmartShunt, battery temp sensing, Class T primary fuse + branch fusing.
- AC protection chain is purchased/locked for Phase 1 (`shore source/adapters -> portable EMS -> shore cord -> L5-30 inlet -> single 6-way AC DIN enclosure -> 30A AC-in breaker -> MultiPlus -> 30A AC-out main -> two 20A GFCI branches`). AC-in/MultiPlus charging has passed a short limited-current live test; AC-out branch/GFCI commissioning remains pending.
- Current build phase: the permanent Lonseal floor has entered controlled module loading. The electrical module is hard-mounted to truck-bed hardpoints and fits cleanly; it still needs the planned Bench tie-in for final anti-rack/road stiffness. The MultiPlus and AC breaker enclosure are temporarily off the backer for easier lifting and will remount into embedded pronged T-nuts. All three `2/0 AWG` battery branch cable sets are cut, lugged, heat-shrunk, and landed at the battery-side busbars, but the batteries remain isolated pending individual charge/rest/voltage matching. Immediate closeout is shore AC-in -> full-bank parallel/commissioning -> `12V` battery/Orion fuse cleanup -> factory lights/Maxxair/DC outlet proof, followed by tank/pump bench testing and installed wet-spine work.

Physical integration snapshot (`2026-07-19`)

- Electrical module is positively hard-mounted through the finished floor to the truck bed. Its present stationary stiffness is acceptable; the Bench tie-in remains part of final mobile restraint.
- MultiPlus and combined AC breaker enclosure were removed before the lift and remain ready to reinstall from the front through embedded pronged/spiked T-nuts in the plywood backer.
- Three `48V` battery branch harnesses are physically complete and landed at the positive/negative battery-side busbars. Battery 2 and Battery 3 still need individual shore charging and a rested-voltage match within `0.1V` before the three batteries are paralleled.
- `12V` buffer branch remains `4 AWG`: `battery + -> F-11 100A ANL -> SW-12V-BATT -> fuse-panel main +`; `battery - -> fuse-panel main -`. Orion output uses `6 AWG` to the same junction through `F-07 60A/80V` on output positive.
- Orion input protection is one verified `40A` MEGA (`58VDC` minimum under the locked `56.8V` charge ceiling; Victron `CIP138040020 40A/80V` is the replacement fallback) in Lynx Slot 4 feeding the existing `6 AWG` pair directly. The 6 AWG input is overkill but already installed and safely protected by 40A; no separate inline or DIN input fuse holder is used. This is an explicit practical deviation from Victron's `20A` external-protection table to eliminate redundant connection stacks. Purchased `30A/58V` MIDI and FKS/ATO stock remains unused. Keep `6 AWG` and `F-07 60A/80V` on the Orion 30A 12V output.

Commissioning snapshot (`2026-05-27`)

- Owner confirmed `55.5V` at the `48V` bus and at the MultiPlus after pre-charge/energization.
- MultiPlus switch `I` brought inverter mode online; inverter light illuminated, slight hum was observed, and no error lights were reported.
- SmartShunt and Orion-Tr Smart are visible in VictronConnect. SmartShunt red fused `Vbatt+` lead should remain battery-side if SOC continuity while the main disconnect is open is desired; the parasitic draw is negligible for normal short storage windows.
- Cerbo GX is powered from the `48V` system through a small inline fused feed and uses `VE.Bus`/RJ45 to the MultiPlus; its Wi-Fi access point/remote-console workflow is active.
- AC input source type should be labeled `Shore power`, not generic `Grid`, for the mobile source-current-limited workflow.
- Household-outlet shore test used a reduced MultiPlus input-current limit (`10A` first test / `12A` maximum policy on a `15A` circuit). Observed values were about `1294W` shore input and about `54.3V x 21.6A` (`~1173W`) battery charge in bulk.
- MultiPlus battery charge profile has been owner-verified by supervised one-battery live behavior: settings were redone, the charger entered bulk, then quickly transitioned to absorption because the battery was already at/near `100%`, as planned. Current target remains absorption/charge voltage `56.8V`, float `54.0V`, short absorption dwell, equalization off, charger current within the connected-battery limit, and source-current limit set for the actual shore outlet. The Dumfume manual's `58.4V` value is documented, but because the same value is also listed as charge-limit/over-charge protection voltage, it is not the active routine target. Leave `DVCC` disabled unless a documented BMS/GX control path is added. No separate second-battery charge test is required just to close the charger-programming gate; AC-out/GFCI and alternator commissioning remain separate gates.
- Normal shutdown/de-energize for the current bench system: MultiPlus `O`, disable Orion if needed, de-energize/unplug shore, wait briefly, then open the main `48V` disconnect. Residual voltage on the Lynx/load side with the disconnect open is expected from device capacitance but must be treated as live until metered near zero.

Modeling rules (procurement-first plus full-load)

- Primary procurement source of truth is `bom/bom_estimated_items.csv`.
- Load model is maintained in `bom/load_model_wh.csv` (model v5) and includes BOM-sourced installed loads, owner-supplied work electronics (kept out of BOM cost totals), and a conservative preliminary/future camper-audio listening profile. Audio is not near-term procurement.
- Legacy workbook WH model assumptions are retired and not used.
- Voltage convention: use `48V` as architecture label, but use `51.2V` nominal for battery Wh accounting.
- Run-length convention: measured physical layout lengths are cut-length source-of-truth; CAD values are planning references only.

Input reference (maintained)

Input	Current value	Source
Battery bank	`3x Dumfume 51.2V 100Ah` (`1S3P`; manual allows up to `1S4P`). Per battery: charge voltage `58.4V +/-0.2V`; recommended charge current `20A`; max continuous charge `100A`; recommended discharge `50A`; max continuous discharge `200A`; over-discharge protect/recover `36.8V/43.2V`; discharge overcurrent `600A`; short-circuit `1800A`; low-temp charge protection approx `41F-50F`, recovery at `50F`; high-temp protection `157F/140F`.	`bom/bom_estimated_items.csv` row 3 + `references/Dumfume_36V_48V_100Ah_Battery_-_User_Manual.pdf`
Inverter/charger	MultiPlus-II `48/3000/35-50`	`bom/bom_estimated_items.csv` row 12
Alternator charging	Dedicated `48V` secondary alternator path (`Mechman + WS500 + APM-48`) with `Upfitter #3 -> WS500 brown ignition` manual control and Lynx Slot 3 alternator branch fuse lock	`bom/bom_estimated_items.csv` rows `168-171`, `176` + `docs/core/ELECTRICAL_48V_ARCHITECTURE.md`
Obsolete pre-Mechman alternator charger/remote	Returned/obsolete; not part of primary layout, fuse planning, or commissioning	`bom/bom_estimated_items.csv` rows `18` and `26`
Legacy single-12V upgrade path	Mechman `370A` + Big 3 path is deprecated under the dual-`48V` migration baseline	`bom/bom_estimated_items.csv` rows `103` and `104`
DC-DC charger	Orion-Tr Smart `48/12 30A` (`360W`); `48V` input is protected by `F-05 40A` MEGA (`>=58VDC`) in Lynx Slot 4 with no second inline fuse, and `12V` output is separately protected by `F-07 60A/80V`	`bom/bom_estimated_items.csv` row 20
12V buffer battery	`12V 100Ah LiFePO4` on shared 12V junction (`F-11` + `SW-12V-BATT`)	`bom/bom_estimated_items.csv` rows 21, 124, and 125

Solar array candidate	Flexible-first placeholder (`~800-1000W` class); prior `9x100W/3S3P` concept is modeling-only and must not drive roof holes, combiner count, or procurement until solar is reopened after shore and alternator charging	`bom/bom_estimated_items.csv` row 24
Solar controller	SmartSolar `MPPT 150/45`	`bom/bom_estimated_items.csv` row 25
Load profiles (BOM + owner-supplied office loads)	`core_workday`, `winter_workday`, `minimal_idle_day`	`bom/load_model_wh.csv`
Owner-supplied office assumptions	Laptop + 27 inch 1440p monitor + tablet/peripheral charging	`bom/load_model_wh.csv` rows marked `Owner-Supplied`

Modeled expected usage

Load totals below are from `bom/load_model_wh.csv` model v5 (BOM loads plus the purchased Ninja SP151 cooking appliance, owner-supplied office loads, and a conservative preliminary/future camper-audio profile).

Scenario	Daily energy	5-day workweek energy	7-day week energy	Dominant contributors
`core_workday`	`3,915 Wh`	`19,575 Wh`	`27,405 Wh`	Laptop, monitor, Starlink, cooking, inverter idle, conservative future-audio allowance
`winter_workday`	`4,829 Wh`	`24,145 Wh`	`33,803 Wh`	Laptop, monitor, Starlink, diesel heater, cooking, conservative future-audio allowance
`minimal_idle_day`	`624 Wh`	`3,120 Wh`	`4,368 Wh`	Fridge + always-on monitoring/detector loads

Capacity analysis (corrected battery bank)

Metric	Formula	Result
Nominal battery energy	`51.2V x 100Ah x 3`	`15,360 Wh`
Usable energy to 20% reserve floor	`15,360 x 0.8`	`12,288 Wh`
Core day depth of discharge	`3,915 / 15,360`	`25.49%` per day
Winter day depth of discharge	`4,829 / 15,360`	`31.44%` per day

Autonomy (no charging)

Scenario	Days (`100% -> 20%`)
`core_workday`	`3.14`
`winter_workday`	`2.54`
`minimal_idle_day`	`19.69`

Charging potential

All values are planning-level and should be replaced with measured charge logs after shakedown tests.

Solar charging (placeholder `~900W` flexible model)

Solar remains deferred until shore charging and alternator charging are working. The prior `900W` / `3S3P` model is only an energy-planning placeholder; final modules, stringing, combiner/fuse count, and roof penetrations stay unlocked. Base planning factor for the target roof setup is `68%` end-to-end harvest efficiency, with sensitivity from `60%` (poor conditions) to `75%` (strong conditions).

Derate component	Planning factor	Note

Nameplate realization	`0.95`	Manufacturing variance and real-world rating spread
Thermal factor (flexible with air gap)	`0.90`	Better than bonded-flat flexible, still hotter than rigid stand-off
MPPT + wiring efficiency	`0.96`	Controller and cable losses
Flat-roof angle/azimuth factor	`0.88`	No seasonal tilt optimization
Soiling + mismatch + partial shading allowance	`0.93`	Dirt, string mismatch, and intermittent shade impacts
Combined planning factor	`~0.68`	Product of factors above

`900W` array efficiency	2 PSH day	4 PSH day	5 PSH day	Net vs `core_workday` at 4 PSH	Net vs `winter_workday` at 4 PSH
`60%`	`1,080 Wh`	`2,160 Wh`	`2,700 Wh`	`-1,755 Wh/day`	`-2,669 Wh/day`
`68%` (planning base)	`1,224 Wh`	`2,448 Wh`	`3,060 Wh`	`-1,467 Wh/day`	`-2,381 Wh/day`
`75%`	`1,350 Wh`	`2,700 Wh`	`3,375 Wh`	`-1,215 Wh/day`	`-2,129 Wh/day`

- Break-even PSH at `900W` :
- `core_workday` (base `68%`): `3,915 / (900 x 0.68) = 6.40` PSH/day.
- `winter_workday` (base `68%`): `4,829 / (900 x 0.68) = 7.89` PSH/day.
- Sensitivity band for `winter_workday` : `7.15` PSH/day (`75%`) to `8.94` PSH/day (`60%`).
- `MPPT 150/45` is not the bottleneck for the current array range.

Alternator charging (dedicated `48V` secondary alternator path)

- Active migration baseline: Mechman dual-alternator kit + WS500 regulator + APM-48 protection module.
- `Lynx Slot 3` alternator branch fuse (`F-04`) is locked at `150A` (`58V/80V MEGA` class) at the house-bank/Lynx end of the alternator positive run. Mechman guidance requires the alternator positive cable fuse within `12 in` of the battery-bank connection; if final layout places the Lynx farther away, add a bank-end fuse holder rather than leaving the branch unfused at the bank end.
- WS500 fused low-current leads are `F-12` (`10A` baseline / `15A` if required for extra-large alternator case) for regulator power and `F-13` (`3A`) for positive voltage sense. Both are treated as `48V` -bank-referenced on this build unless the final harness documentation proves otherwise, so holder/fuse voltage rating must cover the `58.4V` charge-voltage system.
- WS500 current-sense high/low wires are not a separate fuse position in the current Wakespeed manual; route as a twisted low-current sense pair to the selected shunt/current-sense point and keep them away from noise.
- Manual charge-enable/disable path is locked to Ford `Upfitter Switch #3` feeding the WS500 brown ignition/enable wire through local inline fuse `F-15` (`3A` , 12V control circuit).
- `WS500` white `Feature-In` is reserved for future automatic fault-interlock work and is not required in Phase 1.
- Mechanical-only staged driving with the Mechman alternator installed but unwired/electrically disabled is owner/Mechman-confirmed acceptable after the idler/belt/noise check passes; this does not commission alternator charging.
- WS500 rough-in default is regulator near the truck-bed house electrical area so analog shunt/battery-sense wiring stays short; run the alternator-leg wiring forward in separate labeled looms from the `2/0` charge pair.
- Cable decision lock for this pass: reuse existing uncut `2/0` inventory for the alternator charge path (`~20 ft` one-way assumed), with dedicated equal-size negative run.
- Obsolete pre-Mechman alternator-charger hardware is returned/removed from active planning and should not appear in primary fuse/layout decisions.

Shore charging (MultiPlus-II charger path)

- Charger limit from model string: `35A` ; Dumfume manual lists charge voltage as `58.4V +/-0.2V` but also lists charge-limit/over-charge protection at `58.4V` , so current commissioning uses a lower routine target: `56.8V` absorption/charge with `54.0V` float. For the `3x` bank, manual-backed current references are `60A` recommended charge total and `300A` max continuous charge total, so the MultiPlus `35A` maximum is within battery-bank limits.
- Current first-live result: AC-in shore charging was proven at household-outlet limits with MultiPlus input limit reduced (`10A` first test; `12A` policy ceiling on a `15A` circuit). Reported Cerbo values were about `1294W` from shore and about `54.3V x 21.6A` into the battery bank in bulk.
- Supervised charge-profile verification: fixed LiFePO4 settings were redone and live behavior matched the plan on the first battery; charging entered bulk, then quickly transitioned to absorption because the battery was already at/near `100%` . Earlier settled snapshot showed `56.8V` on the MultiPlus/SmartShunt with `0A` charge current, battery display about `55.8V` , SmartShunt SOC set to `100%` , and shore input current limit `12A` on the household-source test.
- Current charger profile: `56.8V` absorption/charge target, `54.0V` float, `52.8V` storage if available, equalization off, lithium temperature compensation behavior disabled/confirmed, minimum absorption time (`1h` in the observed `1-8h` field), repeated absorption `0.25h` , and

charger current at or below the MultiPlus `35A` limit; if charging one isolated battery, cap current at `20A`.
- `DVCC` remains disabled in the current architecture because there is no documented BMS-to-GX control path.
- AC input current policy: label AC Input 1 as `Shore power`; use `10A` for first tests and `12A` maximum on a normal `15A` household circuit; use the actual pedestal/source rating for `20A` / `30A` sources.
- Ideal bulk-only recharge times at the current `56.8V` commissioning target and `35A` charger limit remain reference-only until measured charge logs replace them:
- Replace one `core_workday`: `1.97h`.
- Replace one `winter_workday`: `2.43h`.
- Recharge full `3x` bank from `20%` to `100%`: `6.18h`.
- Recharge one isolated `48V 100Ah` battery from `20%` to `100%`: about `3.61h` at the single-battery `20A` current cap.
- Real-world times are longer due to absorption taper near full charge, reduced household input-current limits, and any configured charge-current derate.

Operational implications and constraints

- Battery capacity now supports roughly `2.5-3.1` office-workdays without charging in the conservative future-audio model, depending on season and reserve policy; actual near-term office-only use may be better.
- With `900W` flexible solar at the base `68%` factor, `4` PSH leaves a material daily deficit for both `core_workday` and `winter_workday`; the future-audio allowance increases that deficit versus the prior office-only model.
- Shore charging can materially recover SOC in a single evening (`~6.18h` from `20%` to `100%` in bulk-ideal terms).
- Alternator recovery potential is expected to materially exceed the obsolete pre-Mechman charger path once the dedicated `48V` alternator path is commissioned.
- Current execution risk is no longer charger-capacity-limited operation; it is migration/commissioning quality (fitment, regulation, protection, and measured thermal behavior).
- MultiPlus-II `48/3000` inverter continuous output (`~2,400W`) can be exceeded by simultaneous induction + Ninja SP151 air fryer/toaster oven + other AC loads, so high-draw AC loads need sequencing.
- Orion-Tr Smart `48->12V 30A` charger (`360W`) is the continuous charging/feed ceiling into the shared 12V junction; buffer battery supports short transients, including audio bass peaks, but sustained overload still requires load budgeting.

Safety baseline

- Positive path sequence: battery -> Class T fuse (near source) -> disconnect -> Lynx Distributor -> fused branch feeds
- Negative path sequence: battery -> SmartShunt -> Lynx Distributor negative bus -> all load returns on load side of shunt
- Dedicated alternator branch grounding rule set: run equal-or-larger dedicated negative cable from secondary alternator to house-bank return path and avoid sheet-metal return paths.
- If the truck uses an RVC ground-sensor loop, route the upgraded ground path through the loop per vehicle requirements.
- Battery thermal strategy: insulated battery box, ducted warm-air branch, thermostat/relay enable logic
- Wiring practices: grommets, loom, glands, abrasion protection, and bend-radius validation
- Reference links from workbook notes:
- `https://youtu.be/dSYKabw_rgs?t=651`
- `https://www.diodeled.com/45-channel.html`

Electrical overview diagram (implementation)

- Full implementation-level topology diagram: `docs/implementation/ELECTRICAL_overview_diagram.md`
- Scope includes fuse IDs, fuse-holder/housing methods, branch wire-gauge selections, and documented sizing assumptions.

RF-003 Distribution Topology Decision (Lynx Locked)

Decision date: `2026-02-12`

Approved architecture for Phase 1:

- `Victron Lynx Distributor M10` (`LYN060102010`) is the single distribution backbone.
- Current tracked unit price is `$192.47`.
- Current Lynx fused outputs in use (`3` total):
- Slot 1: MultiPlus-II `48/3000` (`F-02 125A`)
- Slot 2: SmartSolar `150/45` (`F-03 60A`)
- Slot 3: Dedicated `48V` alternator branch output (`F-04 150A`)
- Slot 4 (`F-05`) feeds Orion-Tr Smart `48/12` through one `40A` MEGA body-marked at least `58VDC`; standalone `F-06` is retired.
- `1x` Lynx Distributor covers the current four active branch outputs with no spare fused slot.
- If another fused `48V` branch is added later, add a second Lynx module or a separately engineered branch in that phase.

Implementation notes:

- `Lynx Distributor` includes the negative busbar, so a separate standalone negative bus is not required in the Lynx path.
- `Lynx Distributor` branch-fuse LEDs need a `Lynx Shunt VE.Can` or `Lynx Smart BMS`; with `SmartShunt`-only monitoring, LED fuse indication is not active.
- Main Class T protection baseline is `3x` (one per battery-positive conductor) and is tracked in the fuse schedule/BOM.

Reference links:

- Lynx Distributor manual/specs: `https://www.victronenergy.com/media/pg/Lynx_Distributor/en/introduction.html`
- Lynx Distributor retail example: `https://www.invertersrus.com/product/victron-lynx-distributor/`

Fuse Determination Baseline (Lynx)

Objective: maintain a complete start-to-finish fuse schedule for the approved Lynx architecture.

Current detailed schedule (active reference):

- `docs/implementation/ELECTRICAL_fuse_schedule.md`

Scope covered:

1. Main battery protection (`Class T`) quantity, rating, and placement.
2. Lynx branch fuses for active outputs: MultiPlus, MPPT, dedicated alternator branch, and Orion input (`F-05 40A`, `>=58VDC`, in Slot 4); no standalone Orion input fuse.
3. WS500 low-current fuse requirements and alternator-branch protection coordination.
4. Any additional protective devices required by manufacturer manuals for both charge-source and load paths.

Method:

1. Pull max current requirements and overcurrent guidance from each device manual.
2. Confirm planned wire gauge/length/insulation assumptions for each run.
3. Size each fuse to protect the conductor first while meeting equipment requirements.
4. Build final fuse matrix with part numbers, quantities, and BOM row mapping.

Solar

- Current direction: flexible-first design path due roof `75 lb` hard panel-weight cap.
- Near-term project policy: keep solar final procurement/string lock deferred deeper into build while preserving routing/passthrough reservations now.
- Hardwall popup wiring baseline for planning: no hidden in-wall solar run; use exterior-rated coiled jumper cable(s) from roof solar exit to a lower-shell weatherproof passthrough.
- BOM scope for that routing method is tracked in `bom/bom_estimated_items.csv` row `121`.
- Open points: exact passthrough location, connector standard (`MC4` direct vs bulkhead adapter strategy), and final flexible module/string configuration before SKU lock.
- Energy takeaway from current modeled load: moving from sub-`600W` class toward `~800-1000W` flexible can materially reduce daily deficit, but modeled office-workday profiles still require charging strategy integration (solar + alternator + shore).

Electrical reference maintenance workflow

- Update trigger conditions:
- Any change to battery, inverter/charger, DC-DC, or solar components in `bom/bom_estimated_items.csv`.
- Any measured field data that materially changes duty-cycle assumptions.
- Any change in appliance selection, owner-supplied office electronics, or expected duty cycle in `bom/load_model_wh.csv`.
- Update process:

1. Update component rows and assumptions in `bom/load_model_wh.csv`.
2. Recalculate scenario daily Wh totals in this file from that CSV.
3. Recalculate autonomy at the active reserve floor (`usable Wh / daily Wh`).
4. Recalculate charging tables with latest charge source ratings and assumptions.
5. Record what changed in `docs/core/TRACKING.md` (decision/risk/open questions) and `logs/LOG.md`.
6. Remove stale assumptions that are not backed by BOM rows or explicit owner-supplied load assumptions.

- Formula quick reference:

```
battery_nominal_wh = battery_voltage_v * battery_capacity_ah * battery_quantity
daily_wh = sum(component_wh_per_day) + conversion_loss_wh
autonomy_days = usable_battery_wh / daily_wh
```

```
solar_daily_wh = array_watts * effective_psh * solar_efficiency_factor
solar_efficiency_factor = nameplate_realization * thermal_factor * mppt_wiring_factor * orientation_factor *
soiling_shading_factor
alternator_daily_wh = dc_dc_output_watts * drive_hours
alternator_practical_w = (alternator_rated_a - vehicle_base_load_a) * alternator_voltage_v * conversion_efficiency
house_charge_current_a = alternator_practical_w / charge_voltage_v
shore_charge_power_w = charge_voltage * charger_current_a
bulk_charge_hours = energy_to_replace_wh / shore_charge_power_w
```

- Retired model note:
- `bom/load_model_wh.csv` v1 (workbook-derived chart), v2 (BOM-only), v3 (BOM plus owner-supplied office loads), and v4 (added preliminary/future camper audio) are superseded by model v5, which adds the purchased Ninja SP151 appliance load while retaining the audio allowance.

Camper audio

- Camper audio implementation owner: [CAMPER_audio_system](). Status: preliminary/future-roadmap, not near-term procurement. Draft package is a DC-first `2.1` camper-only system: Samsung `S11 Ultra` tablet -> Kicker `46KMC2` marine media receiver -> Kicker `CSC67` `4 ohm` speaker pair plus Kicker `49PTRTP10` powered down-firing 10 in subwoofer.
- BOM rows `189-193` track deferred/preliminary source unit, speaker pair, powered sub, 4 AWG sub power/fuse kit, RCA/speaker wiring, mounts, and install consumables. Row `101` is now only a deprecated sound-system placeholder.
- Electrical posture: KMC2 uses a `15A` source/head-unit branch from the `12V` fuse panel; PTRTP10 uses a separate `40A` source fuse near the `12V` source takeoff with `4 AWG` positive and matching `4 AWG` return to the `12V` negative bus/main stud. Audio returns should not use shell/chassis as the normal current path.
- 12V headroom note: the selected audio system can theoretically peak around `55A` at `12V` (`15A` source unit + `40A` powered sub branch), above the Orion `30A` continuous 12V feed. The `12V` buffer battery supports peaks and short loud sessions, but sustained high-volume use should be watched on the `12V` battery voltage/SOC while other 12V loads are running.
- Physical placement default: KMC2 in the driver-side electrical/workstation/DC-shelf face; powered sub low in a dry driver-side toe-kick/step-box or dry bench volume near the `12V` junction; speaker cutouts/pods wait for wall panel thickness, roof-down sweep, and furniture-service checks.

Communications

- Primary internet path: Starlink Standard (DC power conversion item tracked in BOM)
- Secondary/fallback path: TBD (cellular path and carrier diversity not frozen)
- Placement constraint: routing and passthrough location unresolved

Plumbing (if included in phase 1)

- Near-term baseline: complete the cold-water galley and the two-port propane hot-water service interface together. Build around tank, pump, accumulator, faucet/sink, BLUE pressurized cold-out, RED hot-return, drain/graywater path, and accessible service shutoffs.
- Interior layout and wet-spine draft owner: [INTERIOR_furniture_layout_and_galley](). Current direction is a passenger-side lofted fridge/wet-spine exoskeleton: Iceco/fridge raised about `16 in` , pump/accumulator relocated into the owner-measured `6 in` gap between the cooler support and house batteries, rear fill/vent and BLUE/RED hot-water service access, graywater cassette, and no interior water-heater volume. The wet pack requires a nonconductive splash divider and drained leak path toward the bed floor, not reliance on the battery cases as water protection.
- Water capacity: purchased `36 gal` wheel-well tank. Full water mass is about `300 lb` before tank/brackets/plumbing; plan around roughly `320+ lb` installed when full.
- Owner-confirmed tank port geometry (`2026-07-16`): four molded ports on each end, with two visibly large and two smaller ports per end; no obvious/open top port. Exact model and thread sizes remain unconfirmed. Inspect for a membrane-covered top boss and model/logo markings before assuming a new sender hole is required.
- Tank restraint posture: the wheel-well form factor is materially better than a tall/skinny vertical tank because it lowers center of gravity and reduces overturning moment, but plusnut wall brackets should be verified as part of the restraint system rather than assumed as the only crash/off-road load path.
- Pump candidate captured: Shurflo 3.0 GPM class.
- Purchased galley fixture baseline: FORIOUS black pull-out faucet with soap dispenser plus Sarlai black `15 in x 15 in` topmount sink with approx. `11 in x 13 in` interior basin (BOM rows `207-208`). The FORIOUS manual and product diagram confirm separate `3/8 in` hot/cold supply-hose connections. The exact EFIELD two-pack of `1/2 in PEX-B barb x 3/8 in OD compression male` adapters (ASIN `B0C7QBNVG9`) was purchased `2026-07-25` ; seat each faucet hose nut directly on the male compression seat with no PTFE tape.
- Current sink-layout implication: the chosen sink is topmount, so the flange can live on the counter while the bowl passes through a local frame opening; still verify actual cutout, faucet shank/nut/hose clearance, drain/graywater routing, and sink-zone rail/connector position

before final cuts.

- Hot-water decision posture (`2026-07-25`, owner-corrected final): propane only. Selected appliance is the outdoor-use Joolca HOTTAP V2 Essentials (`37,500 BTU/hr`, `0.6-1.5 GPM`, recommended source `40-60 PSI` and `1.6-3.17 GPM`, about `17.7 x 11.4 x 6.6 in`) with its supplied four-foot QCC1 regulator/hose and one Flame King `YSN10LB-ALM` `10 lb` aluminum cylinder. The heater and cylinder both store in the rear swingout box. The heater remains mounted to an articulating arm based inside the box, but its burner body swings fully beyond the box opening before ignition. The camper and box each receive a BLUE/RED two-port plate; two short, sleeved jumper hoses are the only water connections made at camp. Internal box hoses stay attached to the heater. A Joolca-compatible three-way hot splitter keeps the shower hose and RED faucet-return branch connected at the same time. Source/fitting details and the complete inline count live in INTERIOR_furniture_layout_and_galley.
- Rear propane package: the Flame King `YSN10LB-ALM` is `9 x 9 x 19.25 in`, `9.6 lb` empty, and about `19.6 lb` filled. It travels upright, valve closed, and LP-disconnected in the owner-selected ventilated bumper box. Capture the cylinder foot ring in a rigid bolted/welded cradle and use a rated body strap as the primary lateral restraint; thin EPDM pads and a rubber strap are anti-chafe/anti-rattle measures, not the only load path. The reported `19.5 in` shelf opening leaves only `0.25 in` nominal vertical clearance, so prove the real cylinder, shelf, valve, regulator-connector sweep, and pad thickness before drilling. Provide permanent low drainage ventilation plus a fixed high vent and keep the valve/collar accessible.
- HOTTAP deployment: Joolca explicitly permits vehicle mounting and publishes an official swing-out-TV-bracket example. Mount the Joolca quick-release bracket to a metal articulating arm whose base is structurally backed inside the box; give the arm an independent positive travel cradle/latch so hinge friction does not carry road loads. Before ignition, open the box door and swing the heater vertical into free air so the actual unit passes Joolca's `39 in` top, `23.6 in` rear, and `19.7 in` front/side operating-clearance checks. Arm reach is measured from the deployed heater faces to the nearest box/door/camper surface, not from the arm pivot alone. Do not burn the HOTTAP while it remains inside the open-front box; an added fan can help cool a storage bay but does not substitute for outdoor classification or clearances.
- Freeze and winterization strategy (`2026-07-25` final): after every freezing-weather use, close LP, close BLUE, relieve pressure at the active outlet, close RED, disconnect and gravity-drain both short jumpers, then drain the box-side service loops and follow the HOTTAP manual's drain/storage procedure. Keep all four CPC panel bodies straight-through/non-valved and both camper service valves accessible so the exterior stubs drain. Use tank drain/open fixtures and regulated blowout only if needed; no permanent antifreeze pickup or three-valve heater bypass.
- Gravity-fill vent hose correction: owner-measured vent nipple is about `10 mm` OD on the main land and `11 mm` OD at the largest barb/ridge. Specify `10 mm ID` food-grade/potable tube, or `3/8 in ID` as the common inch fallback; the previously ordered `1/2 in ID x 5/8 in OD` tube is oversized.
- Current water-integration order: inventory and measure the eight reported end ports and mock the KUS sender/fill/vent/outlet/drain on the workbench; then perform one bare-tank in-truck dry fit before making any new penetration. The exact `2-3/8 in` / `60 mm` hole saw for the KUS `FLS-U` opening was purchased `2026-07-24`, but owning the tool does not clear the cut gate. Current default is an upper large end port for gravity fill and the highest suitable upper small end port for the unrestricted vent/overflow. The controlling geometry is continuous downhill fill-hose routing from the exterior hatch plus continuous vent rise without a trapped low loop; neither function requires a literal top-surface port if the selected end ports remain in the tank airspace.
- Tank-level electrical baseline: purchased KUS SSS/SSL `14.5 in`, US `240-30 ohm`, two-wire sender connects directly to one native Cerbo GX MK2 Tank-input column. Use black as signal and pink as its paired return over `18-22 AWG` duplex; return to the matching Cerbo negative rather than chassis. The Cerbo originally included the removable Tank terminal block; physically locate it and verify duplex/splice stock before tank closure. No separate 12V feed, fuse, analog gauge, VE.Direct cable, or GX Tank 140 is required.
- Current plumbing-route concept (`2026-07-25` final propane baseline): draw from a low north-end port through the existing tank shutoff, full-flow 90 only if the real turn requires it, flex, strainer, pump, discharge flex, and accumulator. Mount the purchased SHURflo/SEAFLO pack in the measured `6 in` cooler-to-battery gap on a thin vibration-isolated plate supported by extrusion; preserve strainer-bowl removal and point fittings/leak paths away from the batteries behind a nonconductive splash divider. The SEAFLO `SFAT-075-125-01` accumulator has two `1/2 in MNPT` pass-through ports; its two included white swivel fittings terminate in flexible-hose barbs, not `1/2 in PEX-B` crimp barbs. Use the purchased RecPro `30 in` `1/2 in FIP x 1/2 in FIP` braided hose from the pump to one accumulator port and one purchased YVSKM `1/2 in PEX-B barb x 1/2 in female swivel` adapter (four-pack ASIN `B0FDFM97HP`) on the other port; retain three spares. The YVSKM listing describes no-lead brass, but exact potable certification and pressure/temperature documentation remain unverified, so inspect markings/gaskets, fit-check, and cold-pressure-test; Apollo `APXFB1212S` / `CPXFB1212S` remains the fallback. After the accumulator, one `1/2 in` PEX tee branches to faucet cold while the straight leg continues to the rear BLUE service valve and camper-side BLUE QD. The separate camper-side RED QD/service valve feeds one insulated, joint-minimized `1/2 in` PEX line to faucet hot. A matching guarded two-port plate under the bumper box feeds permanently connected BLUE inlet and RED return service loops on the heater arm. Two short, color-keyed, double-ended CPC jumpers connect camper to box only while parked. There is no electric hot-water appliance or dedicated branch, camper manifold, bypass, permanent interior heater branch, or concealed joint behind the tank. Keep the gravity tank drain and sink gray drain physically separate from both pressure-service ports.
- Exterior-cut dependency: the shore inlet waits for the hard-mounted electrical module and proven AC-in cable path; the water-fill/vent hatch waits for final tank orientation, port/fitting method, hose bend/fall, service access, and wet-side backing. Do not let exterior convenience choose an inside route that cannot be serviced.

Cabinetry and structure

- Interior furniture/layout draft owner: [INTERIOR_furniture_layout_and_galley]. Recommended planning direction is the office-first hybrid: passenger-side lofted Iceco/fridge on an extrusion exoskeleton, pump/accumulator below the fridge next to the `36 gal` wheel-well tank, adjacent separated battery bench, driver-side electrical closet/workstation/DC shelf, diesel heater low in the driver-side utility zone, and clear center aisle/cabover movement.
- Driver-side workstation implementation draft: [INTERIOR_driver_side_workstation]. It captures the current best direction for a driver-side service-spine desk, stow-low monitor cassette/rising VESA spine, electrical-closet/DC-shelf interface, shallow storage, travel locks, cable management, heat/noise control, and roof-close acceptance tests.
- Aluminum extrusion strategy: the broad `15-series` starter order is superseded. Default to `10-series` prototype stock/hardware for furniture where practical; reserve larger extrusion for measured heavy/dynamic freestanding modules that prove they need the stiffness, such as the lofted fridge skeleton, electrical closet frame, monitor mast/spine, or battery bench structure if dry-fit proves the load path needs it.
- Panel strategy: living-facing furniture surfaces should use mechanically removable overlay panels over the frame; service zones stay exposed or quick-removable. Magnetic service covers are acceptable only for light non-structural panels and need locator/anti-shear features plus backup retention where loss would matter.
- Current furniture CAD and May 4 generated diagrams are reference-only after installed-shell layout changes; re-CAD block envelopes around the passenger-side lofted fridge/wet-spine, tank overlap, battery bench, driver-side electrical closet, and desk before final cut lists.
- Modular mounting baseline now includes T-slot/strut rails both exterior (recovery/tool mounts like shovel/Maxtrax) and interior (baskets/hooks/tie-down points); BOM rows `119` and `120`.
- Drawer hardware baseline includes `4x` soft-close undermount slide kits for primary cabinetry drawers; BOM row `122`, but final lengths are deferred until fridge/tank/electrical module envelopes are verified.
- Desk concepts captured: fixed full-time work surface with possible auxiliary fold leaf; Lagun-style or pneumatic concepts should remain secondary unless the measured work envelope proves they are stable enough.
- Material ideas captured: phenolic/richlite top, sound treatment, panel anti-rattle tape, frosted/angled LED strips, smoke-grey acrylic shallow cubby covers.
- Monitor travel strategy concept: stow-low/face-down assisted deployment with structural bracing, hard stops, positive travel locks, cable drag chain, and a visible roof-safe state before pop-down.

HVAC and condensation

- Purchased heater baseline: LF Bros `5 kW / 12V / 10 L Split Pro`, with supplied tank, external metering pump/filter, exhaust/muffler, combustion intake, T4S hardwired controller, wireless remote, harness, and mounting hardware. Detailed fit/cut owner: [INTERIOR_driver_side_workstation].
- Heater install posture: horizontal/upright and low in the driver-side utility zone; preserve at least `10 cm / 4 in` at the rear cabin-air inlet and an unrestricted front warm-air path. The underside exhaust, combustion-air intake, fuel inlet, and mounting studs must meet a sealed metal through-floor plate/turret directly below the heater. Do not pass exhaust through the plywood electrical backer or an interior electrical cavity.
- Truck-cut baseline: LF Bros demonstrates two `30 mm` combustion holes plus four stud holes on a simple flat metal surface. Hiatus' layered floor and corrugated bed require the actual metal turret/plate template and an under-truck obstruction check to control the final cut; use the demonstration only as interface evidence, not as the final camper-floor cut pattern.
- Outside routing baseline: exhaust and muffler stay completely outside, fall slightly for condensate, use broad sweeps rather than a hard `90-degree` turn at the heater, and remain separated/shielded from fuel, wiring, plastic, undercoating, openings, and the combustion intake. Intake and exhaust must not terminate in the same direction or where exhaust can be recirculated.
- Supplied tank/pump baseline: the pump does not enter the tank. Drill a centered `7-8 mm` hole through the molded lower tank-outlet boss for the supplied pull-through bulkhead/nozzle; do not cut out the full apparent `3/4-1 in` pad. Fuel order is `tank -> filter -> pump -> heater`; pump outlet points toward the heater and upward about `45 degrees`, with pump-to-heater distance within `2 m`.
- Fuel-storage posture: do not leave the open plastic tank/fill/filter/pump in the battery/electrical/living bay. Preferred direction remains exterior mounting or sealed externally vented containment with spill control, with only a protected fuel line entering the heater zone. A larger narrow exterior tank remains optional later; the supplied `10 L` tank and pump already cover commissioning hardware.
- Electrical/control baseline: the harness branches to heater, pump, T4S controller, and `12V +/-`; the pump is ECU-controlled and the wireless remote needs no wire or cut. Feed from fused `12V C-22`, never `48V`, and preserve power throughout controller-commanded cooldown. Measure the final route before accepting the planned short-run `14 AWG / 15A` branch; use `12 AWG` if length or startup-voltage testing demands it.
- Ventilation: Maxxair fan included in current camper config
- Lighting split: Hiatus factory overhead LED+dimmer remains separate from the planned ambient/cabinet LED subsystem. Desired future interior-lighting design is `12V` QuinLED/WLED analog PWM control on the dedicated lighting branch, with upper CCT white wash, lower RGBCCT night/entry strip, and hardwired momentary buttons; install/procurement is deferred.
- Condensation controls and climate envelope limits: TBD

Safety

- Purpose: define a practical, build-ready safety baseline for the current architecture (`48V 15.36kWh` house bank, `12V` distribution, `120VAC` shore/inverter path, and propane-supported heating/hot-water concepts).
- Priority order: prevent ignition and overcurrent faults, preserve safe shutdown paths, detect hazards early, and make isolation/service repeatable.
- Final install gate: before energizing or using propane in service, verify all items against manufacturer manuals and complete licensed inspection where required.

System-wide controls

- Keep one-line diagrams, fuse IDs, and conductor IDs synchronized across:
- `docs/implementation/ELECTRICAL_overview_diagram.md`
- `docs/implementation/ELECTRICAL_fuse_schedule.md`
- `docs/core/TRACKING.md`
- Ensure all protection and isolation devices are physically accessible without disassembling fixed furniture.
- Keep gas components and AC/DC electrical components separated by design; no mixed service cavities without physical barriers and clear labeling.
- Label every branch and shutoff point so an operator can isolate faults quickly under stress.

48V battery system safety (primary)

- Main hazards: high fault current, sustained DC arc potential, short-circuit heating, incorrect polarity during service, and thermal stress events.
- Required architecture controls:
- Battery positive path stays: battery -> Class T fuse near source -> main disconnect -> Lynx bus; load/charge branches are protected by Lynx fused outputs or, for Orion input, standalone source-side `F-06` at the Lynx bus tap.
- Battery negative path stays: battery -> SmartShunt -> Lynx negative bus (all returns on load side of shunt).
- Use only voltage-appropriate overcurrent devices on house DC branches (`58V` / `80V` class on `48V` paths); do not substitute `32V` automotive-only fuses on `48V` circuits.
- Keep the Orion `F-06` source-side tap from the Lynx bus physically short and protected; do not leave a long unfused lead between the Lynx bus and the inline fuse.
- Keep all busbars/studs covered and insulated; use boot covers, strain relief, and abrasion protection on all near-bus runs.
- Manual alternator shutdown order stays: `Upfitter #3 OFF` first to disable the `WS500`, then open the main `48V` disconnect only after alternator charging is no longer active.
- Commissioning controls (first energization and after major rework):

1. Verify polarity and expected voltage at each segment before inserting branch fuses.
2. Confirm torque marks on all high-current terminals and re-check after initial thermal cycles.
3. Confirm the disconnect fully de-energizes downstream service zones as intended.
4. Validate no unintended parallel return paths bypassing shunt measurement.

- Service controls:
- Remove conductive jewelry, use insulated tools, and keep one-hand work practice on live-exposure checks.
- Never perform branch rewiring with battery disconnect closed unless the specific test requires energized state and a spotter is present.

12V distribution safety

- Main hazards: feeder overload from Orion output limits, hidden voltage drop causing heat at terminations, and unfused accessory additions.
- Required controls:
- Keep Orion output feeder fused at source (`F-07`) and avoid adding unfused taps between Orion and the shared 12V junction.
- Keep 12V buffer battery positive protected at source (`F-11`) and route service isolation through `SW-12V-BATT` downstream of `F-11`.
- Keep `SW-12V-BATT` in its normal closed position during operation; use open position only for service isolation/diagnostics.
- In normal closed operation, Orion supports both active 12V loads and buffer-battery maintenance through the shared junction path.
- In this baseline, the 12V fuse block is the shared junction device (`main +` stud combine point plus integrated negative bus return point).
- Do not solder-splice high-current 12V source conductors; use crimped lugs on rated stud terminals.
- Maintain branch-level fuse-to-conductor coordination per `docs/implementation/ELECTRICAL_fuse_schedule.md`.
- Keep always-on detector branch (`12V-05`) protected but never switch-controlled.
- Keep ambient/cabinet strip lighting on the dedicated DC branch (`12V-11`) so low-light use does not require inverter operation. Desired future path is `12V` QuinLED/WLED analog PWM control, not the superseded `24V` converter/MiBoxer worksheet; verify final fuse/conductor sizing once selected strip wattage and expansion channels are known.

- If sustained `12V` demand exceeds Orion headroom, treat additional `48V->12V` charger capacity (`BOM row 118`) as a safety action, not a convenience upgrade.

120VAC shore/inverter safety

- Main hazards: shock from miswired neutral/ground, ground-fault exposure at outlets, and overcurrent heating from undersized branch protection.
- Required controls:
- Keep shore path order: shore source/adapters -> portable EMS/surge protection -> shore cord -> shore inlet (`L5-30`) -> combined AC DIN enclosure (`30A` UL489 AC-in breaker/disconnect) -> MultiPlus AC-in.
- Keep AC-out path order: MultiPlus AC-out-1 -> `10/3` feeder -> combined AC DIN enclosure (`30A` AC-out main + `20A`/`20A` branch breakers) -> GFCI receptacle on each active branch.
- Keep AC-in hardware on a `30A` / `10 AWG` basis, and set MultiPlus input current limit to the actual source (`15A`, `20A`, or `30A`) whenever adapters are used.
- Keep AC-out-2 as reserve-only in Phase 1 (labeled capped route; no energized branch hardware yet).
- Preserve continuous equipment grounding and chassis bond through all AC paths.
- Do not add a fixed downstream neutral-ground bond in branch wiring; neutral/ground behavior must follow MultiPlus transfer/bonding design.
- Commissioning checks:

1. AC-in-only charger validation with AC-out loads disconnected for first battery charge.
2. Receptacle polarity test on each outlet location.
3. GFCI/RCD trip test at each protected branch.
4. Verify AC-out-2 remains de-energized as a reserve-only capped route in Phase 1.

Propane safety (rear-mounted tank plus passthrough and hot-water path)

- Main hazards: leak accumulation, ignition near electrical equipment, CO exposure, and using outdoor-only appliances in enclosed space.
- Rear-mount tank controls:
- Keep cylinder upright and externally mounted with impact-resistant bracketry and valve protection.
- Keep tank shutoff valve accessible from outside without tools.
- Protect hose/regulator routing from road debris, abrasion, and exhaust heat zones.
- Propane passthrough controls:
- Use sealed bulkhead/pass-through hardware with chafe protection at all penetrations.
- Avoid concealed unions/joints in inaccessible cavities; keep serviceable connections at inspection points.
- Perform leak tests after any connection change and before each trip phase where propane is used.
- Appliance selection rule (critical):
- Treat portable outdoor tankless heaters as outdoor-use-only unless a specific model is explicitly listed for indoor/RV enclosed installation with compliant venting.
- If an indoor propane water-heating path is pursued, lock to an RV/marine-listed indoor unit with approved venting/combustion-air method and documented install clearances before purchase.
- Detection and ventilation controls:
- Keep an LP detector low in cabin, CO detector in breathing zone, and smoke detector high in cabin.
- Test detector alarm functions on a recurring schedule and replace by manufacturer expiration date.

Emergency shutdown baseline

1. Remove active high-draw loads (AC and gas appliances) if safe to do so.
2. Open the `48V` main disconnect and remove shore input.
3. Close propane cylinder valve at the tank.
4. Ventilate cabin area and confirm detectors are active.
5. Use fire extinguisher only if contained/incipient and exit path is clear; otherwise evacuate and call emergency services.
6. Do not re-energize or reopen gas supply until fault root cause is identified and corrected.

Safety hold points before walls/panels are closed

- Complete and document high-current DC inspection (polarity, fuse value, terminal torque, insulation/boots).
- Complete AC verification (polarity, GFCI/RCD trip tests, branch labeling, ground continuity).
- Complete propane leak check and pressure-hold verification with all planned valves/fittings in final positions.
- Complete detector placement and functional alarm tests.
- Record evidence in `logs/LOG.md` and track unresolved items in `docs/core/TRACKING.md`.

Source artifacts

- `docs/legacy/SYSTEMS_workbook_build_notes_obsolete.md`

- `bom/load_model_wh.csv`